

0 3

Farmers use artificial fertilisers to maintain or increase yield from grain-producing crop plants such as wheat.

0 3 . 1

Artificial fertiliser is used to replace mineral ions removed from the land when crops are harvested. One of the mineral ions is nitrate.

Give **two** examples of biological molecules containing nitrogen that would be removed when a crop is harvested.

[2 marks]

1 _____

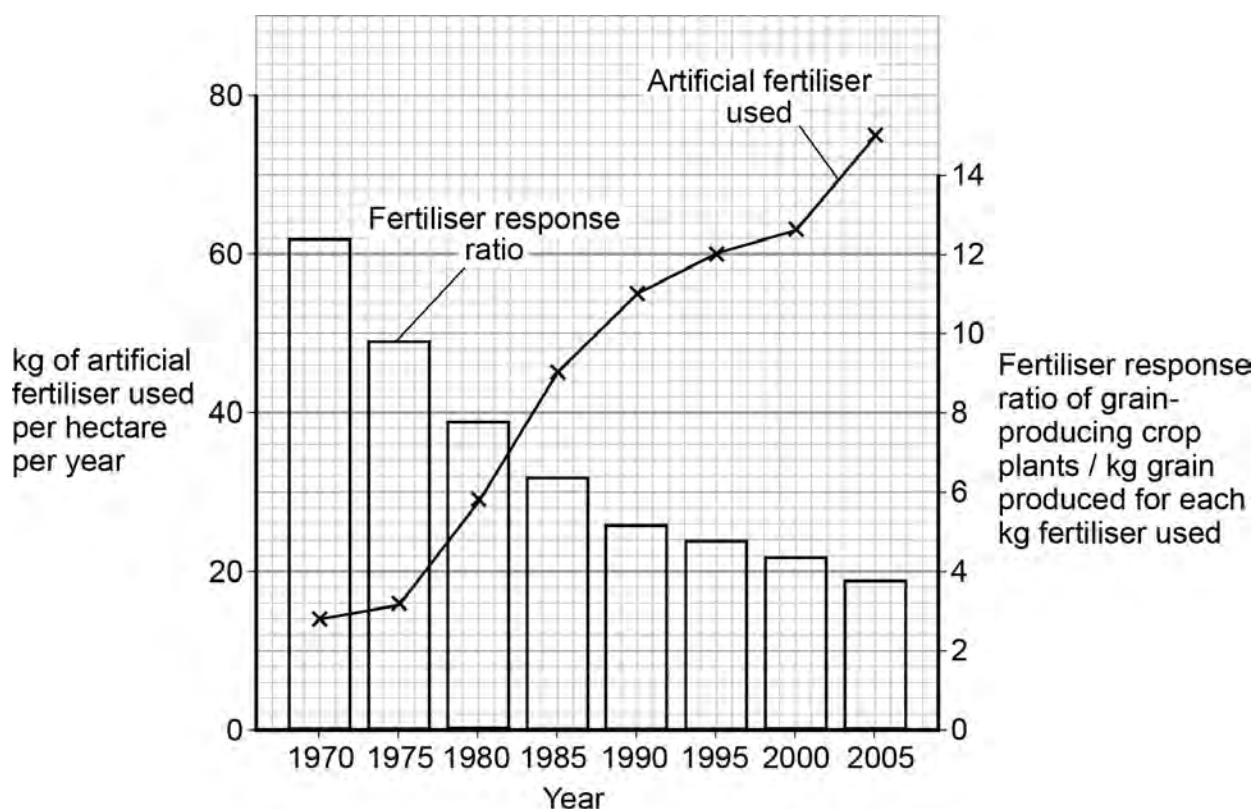
2 _____

0 3 . 2

Scientists investigated changes in the use of artificial fertiliser in India between 1970 and 2005. They also investigated changes in the **fertiliser response ratio**. This ratio shows how many kg of grain are produced for each kg of fertiliser used.

Figure 4 shows their results in the form the scientists presented them.
(A hectare is a unit of area commonly used in agriculture)

Figure 4



Use these data to calculate the difference in the mass of grain produced per hectare in 1970 compared with 2005.

Show your working.

[2 marks]

Difference _____ kg hectare⁻¹

0 3 . 3

Use the data in **Figure 4** to evaluate the use of artificial fertilisers on grain-producing crops in India.

[2 marks]

Turn over for the next question

Turn over ►

